Paleomagnetic results from the upper Keweenawan Chequamegon Sandstone: implications for red bed diagenesis and Late Precambrian apparent polar wander of North America

CHAD MCCABE AND ROB VAN DER VOO
Department of Geological Sciences, The University of Michigan, Ann Arbor, MI 48109, U.S.A.

Received May 4, 1982
Revision accepted August 18, 1982

Two components of magnetization have been isolated from Chequamegon Sandstone samples using chemical demagnetization. A characteristic magnetization resides in magnetite of detrital origin. The pole calculated from this magnetization is 12.3°S, 177.7°E (K = 111.5, A_{95} = 4.6°). This pole lies with other poles of late Keweenawan age and is very close to the Jacobsville Sandstone poles. A secondary magnetization resides in authigenic hematite and yields a pole close to the present north pole. This high-latitude pole is known from an earlier study of the Chequamegon and has been used as evidence for the Hadrynian APW track. However, most if not all of the high-latitude poles that define the Hadrynian track are secondary and undated. Our preferred alternative to the Hadrynian track is that the high-latitude poles are recent remagnetizations and that the antipodal equatorial poles that mark its end points represent field reversals.

Deux composantes d'aimantation ont été résolues à partir d'échantillons du grès de Chequamegon au moyen d'une désaimantation chimique. Une aimantation caractéristique est portée par la magnétite d'origine détritique. Le pôle calculé à partir de cette aimantation est 12,3°S, 177,7°E (K = 111,5, A_{95} = 4,6°). La position de ce pôle est analogue à celle d'autres pôles d'âge du Keweenawan supérieur et elle coincide d'assez près avec les positions des pôles du grès de Jacobsville. Une aimantation secondaire est portée par l'hématite authigénique et fournit un pôle voisin du pôle nord actuel. Ce pôle de latitude élevée a été rapporté dans une étude antérieure du Chequamegon et a été utilisé pour tracer la courbe de la dérive apparente des pôles à l'Hadrynien. Cependant la plupart, sinon la totalité des pôles de latitude élevée qui déterminent la courbe à l'Hadrynien sont secondaires et non-datés. Nous préférons à la place de la courbe à l'Hadrynien, une interprétation attribuant les pôles de latitude élevée à des réaimantations récentes et que les pôles antipodes équatoriaux qui constituent les termes extrêmes représentent des inversions de champ.

[Traduit par le journal]

Can. J. Earth Sci., **20**, 105–112 (1983)

Introduction

Several Late Proterozoic apparent polar wander (APW) trends for the North American craton have become well established in recent years. Particularly well known are the Logan Loop (Robertson and Fahrig 1971) and the Grenville Loop (McWilliams and Dunlop 1975). The Logan Loop is a broad APW loop that has been defined from numerous studies in the Lake Superior region, the Grand Canyon (Elston and Grommé 1974), and other areas of the craton. Asymmetric reversals have been observed in the lower and middle Keweenawan age rocks that define the northern apex of the Logan Loop. Much careful work has been done in order to explain the origin of these asymmetric reversals (e.g., Palmer *et al.* 1981; Halls and Palmer 1981; Pesonen and Nevanlinna 1981) and they may be due to non-dipole behavior of the Earth's magnetic field during this interval. Based on such a non-dipole model, Pesonen and Nevanlinna have offered a modified Logan Loop that is greatly reduced in size from that originally proposed. The descending limb of the Logan Loop is defined by poles from middle and upper Keweenawan volcanics and sediments. A critical link between the Keweenawan poles and the later Grenville poles is provided by secondary poles from the Middle River section (Orienta Formation) and the Fond du Lac Sandstone (Watts 1981).

The Grenville poles, once considered as evidence for convergence between the Grenville Province and the North American craton, have been shown to represent magnetizations reset during the cooling of the Grenville Province (McWilliams and Dunlop 1975). Thus they postdate the orogenic event and must be considered cratonic North American poles.

In addition to the well established Keweenawan and Grenville APW path segments, some poorly dated magnetizations, many demonstrably secondary, have been used to define an APW path termed the "Hadrynian track" (Morris and Roy 1977). The Hadrynian track involves major APW that begins with the 950 Ma Grenville poles, moves over the present north pole, and terminates about 180° away from the Grenville poles in the equatorial Atlantic Ocean with poles dated at 670 Ma. Since the high-latitude poles that define the Hadrynian track are undated, time calibration of this APW path segment is controlled only at its beginning

0008-4077/83/010105-08$01.00/0
©1983 National Research Council of Canada/Conseil national de recherches du Canada

and end, spanning the interval 950–670 Ma. Roy and Robertson (1978) have discussed several possibilities for the age of the undated high-latitude poles, including that they are of recent origin, but prefer a Late Precambrian age.

Piper (1981) has reported results from the west Greenland kimberlite–lamphrophyre suite and has proposed a "Hadrynian" APW path with an entirely younger age than that of Morris and Roy (1977). Radiometric age determinations for the Greenland rocks are bimodal and fall about at 1230 and 600 Ma. However, paleomagnetic sampling was not systematically tied to dated units. The site-mean paleomagnetic directions are westerly and form a streaking distribution with shallow to steeply positive inclinations. A shallow southerly cluster was also observed. Piper argued that the streak of directions is the result of APW, though we would suggest (as also noted by him) that the results can be interpreted as the simultaneous removal of a primary magnetization and a secondary overprint. His results could also reflect unobserved tectonic tilting and rotation of these intrusives. Despite the ambiguity of the age and interpretation of these magnetizations, Piper used his data to construct a revised Hadrynian track, similar in shape and position to that proposed by Morris and Roy (1977) but constrained in time between about 600 and 540 Ma. In Piper's model the velocity of the North American continent with respect to the pole would be about 30 cm/year, an extremely high rate compared with those known for the Phanerozoic.

Since the high-latitude poles that define these APW paths are undated, doubt must remain concerning their validity. Nevertheless, Piper (1981) used his North American APW path to propose a particular reconstruction of a Late Precambrian supercontinent. The importance of such reconstructions in global tectonic studies warrants the close examination of the poles that at present define these APW paths.

One critical high-latitude pole in both Hadrynian track models is the one derived from the Chequamegon Sandstone (DuBois 1962). This pole was calculated from a small number of samples that were subjected to minimal thermal cleaning. The present study was undertaken to apply modern paleomagnetic methods to test whether this earlier Chequamegon pole is the result of a primary or a secondary magnetization.

Geologic setting

The Chequamegon Sandstone is the uppermost unit of the Bayfield Group. It lies near the axis of the Keweenaw trough in horizontal beds that crop out on the Bayfield Peninsula and adjacent Apostle Islands in Wisconsin (Fig. 1). Bayfield Group clastics below the Chequamegon are the Devils Island Sandstone, the Orienta Sandstone, and their probable equivalents. Sedimentary structures suggest unimodal easterly current directions and that the Bayfield clastics were deposited by fluvial processes (Myers 1971). The Bayfield Group overlies the Oronto Group; together these two groups comprise upper Keweenawan sediments, which overlie Middle Keweenawan volcanics that are generally thought to have been extruded at about 1100 Ma (Irving and McGlynn 1976).

The age of the Bayfield Group and its probable equivalent in Michigan, the Jacobsville Sandstone, is constrained to be younger than middle Keweenawan and older than overlying Upper Cambrian sediments. Paleomagnetic results from the Jacobsville (Roy and Robertson 1978) show that the Jacobsville pole falls on the APW path in agreement with a late Keweenawan age.

The major structure in the area is the Keweenaw trough, which appears to underlie most of Lake Superior (Halls and West 1971). This trough continues to the southwest under Paleozoic cover and is aligned with the mid-continent gravity high. The Keweenaw trough is considered to represent a failed intracontinental rift (Chase and Gilmer 1973). The Bayfield sandstones are bounded to the south by the Douglas Fault, a high-angle reverse fault that puts these sandstones in fault contact with middle Keweenawan volcanics.

Paleomagnetic sampling and results

A small pilot collection of Chequamegon and Devils Island samples, collected by D. Watts, was studied to test the suitability of these units for a paleomagnetic investigation. The samples were demagnetized in the laboratory using thermal, alternating field (AF), and chemical demagnetization techniques as well as combinations of the three methods. Thermal demagnetization to 680°C and AF demagnetization to 100 mT appeared not to resolve the component magnetizations satisfactorily. Only chemical leaching in HCl to near complete removal of the natural remanent magnetization (NRM) enabled isolation of the characteristic magnetizations while producing consistently good results.

A more extensive collection of Chequamegon Sandstone samples was taken during the 1980 field season. From 10 sites on the Apostle Islands and the Bayfield Peninsula 149 samples were taken. Samples were drilled in the field with a portable rock drill and oriented with a Brunton compass and clinometer.

All samples from this second collection were slotted to improve leaching uniformity (Henry 1979) and demagnetized in the laboratory by immersing them in 8 *N* HCl and allowing them to leach while in a near-zero field environment. At intervals the samples were rinsed in distilled water and measured with an ScT cryogenic magnetometer. Each time the samples were measured fresh acid was prepared for the next treatment. After

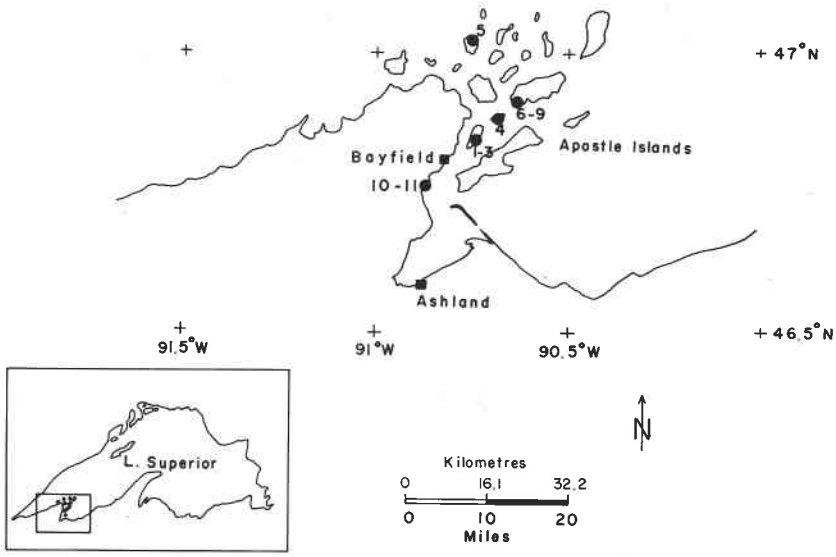


FIG. 1. Map of the study area showing the location of sites.

approximately 100 cumulative hours of leaching, treatments in 8 *N* HCl became less effective and the acid concentration was increased to 10 *N*.

A few samples were removed from the treatment process before leaching had gone to completion and subjected to stepwise AF demagnetization. In most cases AF demagnetization removed a significant portion of the remaining NRM. However, decay in alternating fields was noisy and leaching to near complete removal of the NRM was judged the superior method for the determination of the characteristic magnetizations.

Chemical leaching resolved two components of magnetization in most samples. However, we observed only one of these directions in some samples, and a few samples showed noisy, uninterpretable behavior. The first component was removed by about 50–150 h of leaching. This direction was observed in 112 of 132 samples that survived the leaching process unbroken. In most cases this component was the dominant component of the total NRM. The site-mean directions cluster extremely well and they are northerly and steeply down. The pole lies at 83.6°N, 84.5°W. Statistical parameters (Fisher 1953) are K = 207.2 and A_{95} = 3.4°. This pole is very close to the present north pole and agrees with the pole obtained by DuBois (1962). This magnetization is interpreted as secondary.

A characteristic magnetization was resolved after about 50–150 h of treatment and decayed toward the origin of orthogonal demagnetization diagrams (Zijderveld 1967). It was observed in 71 of the 132 samples and the direction is easterly and shallow down. The resulting pole is located at 12.3°S, 177.7°E with K = 111.5 and A_{95} = 4.6°. This pole is similar to other late Keweenawan poles. Representative demagnetization diagrams are shown in Fig. 2 and directions are plotted in Fig. 3. Site paleomagnetic data are listed in Table 1.

Rock magnetic results

As noted earlier, AF demagnetization of partially leached samples removed a significant portion of the remaining NRM. This shows that the characteristic direction is carried, at least in part, by a low-coercivity carrier such as magnetite. In order to test for the presence of magnetite, rock magnetic experiments were performed.

The acquisition of isothermal remanent magnetization (IRM) was studied in both untreated samples and in those that had been leached to the point that the characteristic direction was in the process of being removed. The results are shown in Fig. 4. Untreated samples show an initial rise in magnetization as the applied field is increased followed by a more gradual rise. Saturation is not reached even in fields as high as 5 T. This behavior suggests that a high-coercivity material, most likely hematite, is present in significant amounts. Leached samples, on the other hand, show saturation by 300 mT, which shows that a low-coercivity material, perhaps magnetite, is the dominant magnetic phase present.

To test further for the presence of magnetite, low-temperature experiments were performed to determine if the magnetite magnetocrystalline transition at about −155°C is present. Leached samples were cooled in liquid nitrogen, given a near-saturation IRM, and allowed to warm slowly while the magnetization was measured in the cryogenic magnetometer. A plot of

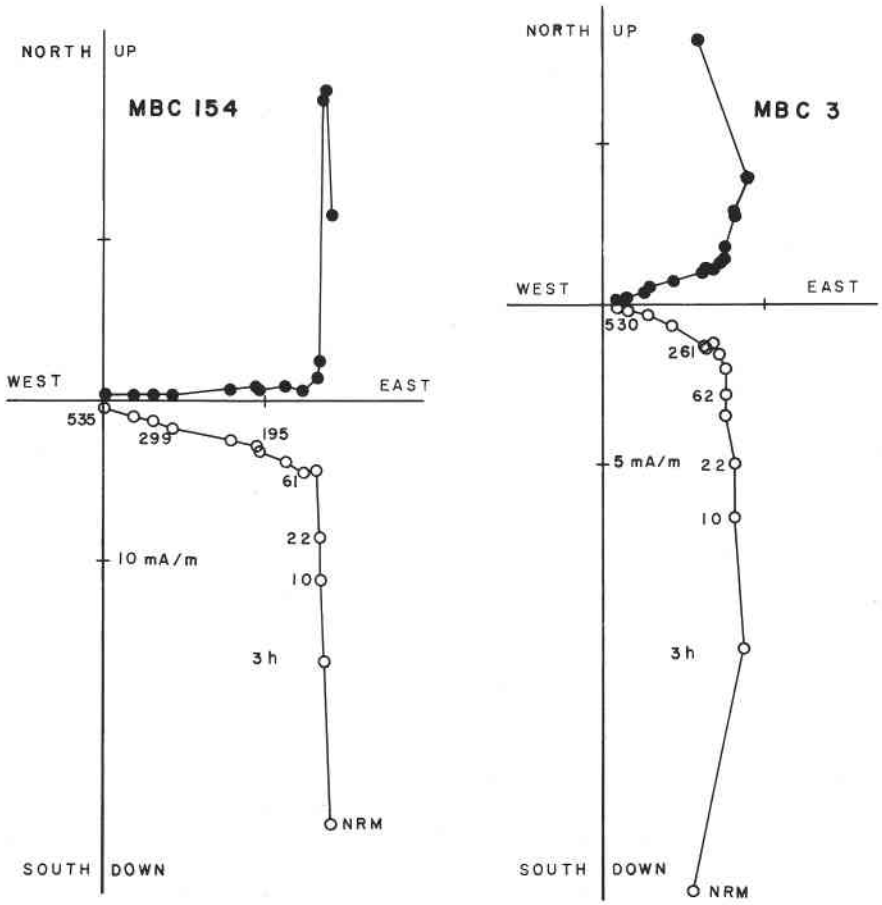


FIG. 2. Representative demagnetization diagrams showing the results of chemical demagnetization. The end points of the magnetization vector during progressive leaching in HCl are plotted. The closed circles are projections on the horizontal plane and the open circles are projections on the east–west vertical plane.

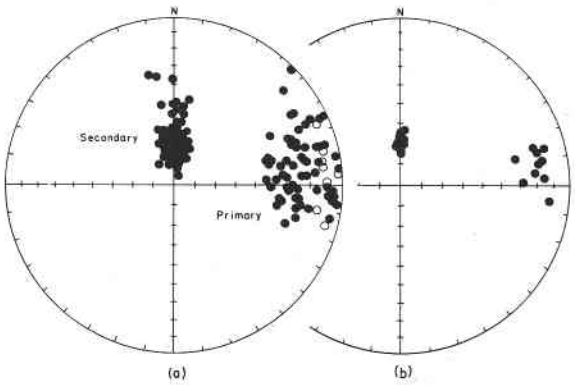


FIG. 3. (*a*) Sample directions plotted on an equal-area net. Secondary and characteristic directions are plotted on the same net as indicated. Solid circles are on the lower hemisphere and open circles are on the upper hemisphere. (*b*) Site-mean directions plotted as in (*a*).

the temperature versus magnetization is shown in Fig. 5. The transition is clearly present, thus confirming the presence of magnetite.

Petrography

Myers (1971) examined a large number of Chequamegon Sandstone samples and found them to be a texturally mature feldspathic quartz arenite. Microcline is the most common feldspar present; rock fragments are mostly volcanic in origin. Some biotite, muscovite, and opaque grains are also present. Grain size is variable but is most commonly medium sand. The matrix is mostly clay minerals with some clay- and silt-sized grains of the same composition as the larger clasts. Red hematite pigment coats both grains and matrix material. Quartz cement is present in some samples, whereas others are rather poorly indurated and friable. Grain alteration is not significant and there is no evidence that the rock has been deeply buried.

TABLE 1. Summary of site-mean paleomagnetic data

	Characteristic direction					Secondary direction				
Site	N/N_0	D/I (°)	k	α_{95}(°)	Pole (°)	N/N_0	D/I (°)	k	α_{95}(°)	Pole (°)
1	10/13	77.6/30.8	29.7	9.0	20.4S, 176.4E	13/13	359.6/66.8	28.7	7.9	87.5N, 96.2W
2	5/13	89.2/27.7	29.7	14.3	11.2S, 169.5E	12/13	2.0/68.5	46.3	6.5	84.6N, 77.8W
3	9/14	81.8/18.3	22.7	11.0	12.5S, 178.7E	13/14	2.9/70.1	121.3	3.8	82.4N, 77.6W
4	6/14	80.5/15.7	14.6	18.2	12.3S, 180.7E	12/14	358.7/69.8	82.7	5.0	83.2N, 97.1W
5*										
6	8/13	87.5/15.6	21.7	12.2	7.5S, 175.8E	8/13	0.9/74.0	370.6	2.9	76.8N, 88.8W
7	6/14	77.0/18.1	10.2	22.0	15.6S, 182.5E	13/14	353.7/71.5	326.8	2.3	80.0N, 111.0W
8	7/14	96.5/11.8	14.7	16.3	0.1N, 170.6E	10/14	0.7/75.2	227.8	3.2	74.8N, 89.3W
9	4/13	74.6/19.3	24.7	18.8	17.8S, 183.8E	9/13	5.3/63.6	34.1	8.9	85.9N, 22.3E
10	7/11	75.9/13.0	10.0	20.0	14.4S, 184.9E	10/11	5.6/67.9	334.6	2.6	84.5N, 50.7W
11	9/13	85.2/21.0	17.9	12.5	11.2S, 175.0E	12/13	0.5/65.7	69.0	5.3	88.8N, 76.3W
Mean poles					12.3S, 177.7E K = 111.5 A_{95} = 4.6°					83.6N, 84.5W K = 207.2 A_{95} = 3.4°

NOTES: N/N_0 is the ratio of samples in which a direction was observed to the number of samples treated; D and I are the declination and inclination of the paleomagnetic direction; k, K, α_{95}, and A_{95} are statistical parameters (Fisher 1953); the mean poles were calculated from site-mean poles.

*Samples were too friable to demagnetize chemically.

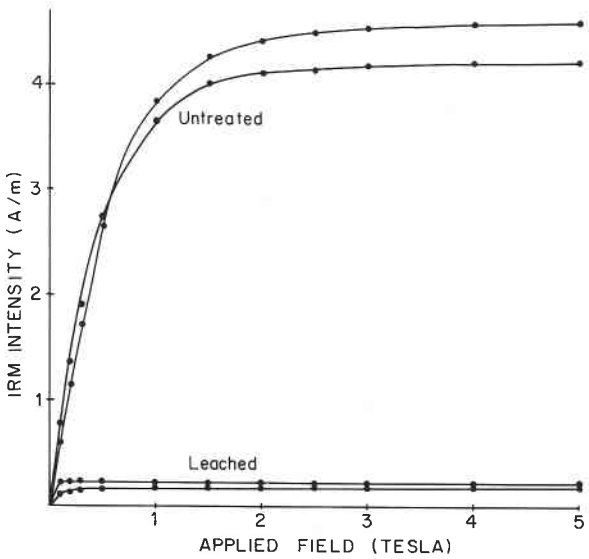


FIG. 4. Acquisition of IRM behavior of two untreated samples and two samples that have been leached in HCl.

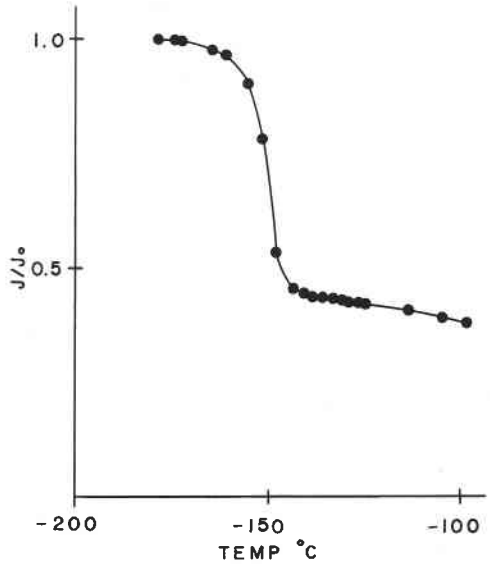


FIG. 5. Low-temperature behavior of a leached sample (see text). J/J_0 is the magnetization normalized to unity.

Fractured surfaces of unleached samples were examined by scanning electron microscopy (SEM) to examine the hematite pigment. Hematite is present as delicate micron-sized plates, identical to the authigenic hematite reported by Walker *et al.* (1981) and Elmore and Van der Voo (1982). Euhedral "books" of a clay mineral, probably authigenic kaolinite, were also observed. Microcrystalline hematite occurs on these clays and it can be concluded that the hematite postdates the formation of the authigenic clay minerals.

Polished thin sections were prepared for reflected light examination of opaque grains, the population of which is dominated by detrital magnetite. Though most have an outer rind that has been altered to hematite, martitization has not been significant, a surprising observation in light of other studies that have shown that

magnetite is usually not common in older red sediments (Van Houten 1968).

Leached samples in transmitted light do not show the conspicuous red staining present in untreated samples, suggesting that leaching removes most of the hematite pigment. Detrital opaque grains became corroded and embayed. Leaching was found to be uniform through the samples, in contrast to the uneven leaching reported by Elmore and Van der Voo (1982) in their study of the Copper Harbor Conglomerate. This difference in behavior may be attributable to greater permeability of the less well indurated Chequamegon Sandstone. With SEM techniques the matrix material can be seen to have become chaotic, and no microcrystalline hematite was found.

Discussion

The low coercivity of the characteristic remanence and the presence of the low-temperature magnetocrystalline transition show that this remanence is carried, at least in part, by magnetite. Further, the presence of detrital magnetite in polished thin sections suggests that the characteristic direction is a detrital magnetization. Thus there is the possibility of several types of error in the direction obtained (see review by Verosub 1977). Despite the risk that the paleomagnetic pole obtained from the Chequamegon Sandstone may not accurately reflect the true paleomagnetic field, the pole is of significance. Its position (Fig. 6) is more in line with the late Keweenawan poles than the Grenville or younger poles. This provides evidence that the Bayfield Group, which is not well dated, was deposited during Keweenawan time and is not, as has been argued, Cambrian (e.g., Hamblin 1958). In addition, the closeness of the Chequamegon pole to those from the Jacobsville Formation support the often stated view that the Bayfield and Jacobsville Sandstones are equivalent units. We note, however, that the Chequamegon characteristic directions are of reversed polarity and that most of the Jacobsville directions are of normal polarity (Roy and Robertson 1978). From a pilot collection of Devils Island Sandstone we have obtained a few directions that are of normal polarity. Thus a polarity transition has occurred between Devils Island and Chequamegon time. The fact that the Jacobsville normal and the Chequamegon reversed poles are so similar may indicate that the earlier period of asymmetric reversals had ceased by Bayfield time.

The most significant aspect of our results lies in the fact that a typically Keweenawan primary pole was obtained. As noted earlier, the high-latitude pole obtained by DuBois (1962) was the only pole previously known from the Chequamegon. The present study has shown that the primary remanence in Chequamegon

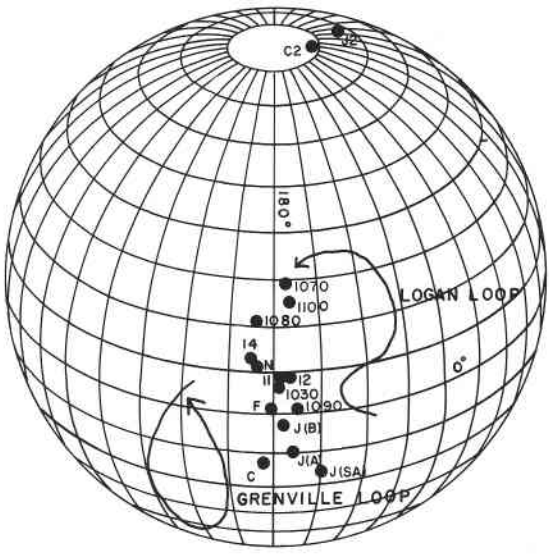


FIG. 6. Some Late Precambrian APW trends for North America. The Logan Loop (as modified by Pesonen and Nevanlinna 1981) and the Grenville Loop (McWilliams and Dunlop 1975) are shown in their approximate positions for reference. Some representative Keweenawan age poles are plotted for comparison with the results of this study. Numbered poles are cited in Henry *et al.* (1977). Freda Sandstone (F) and Nonesuch Shale (N) poles are also from Henry *et al.* (1977). Jacobsville poles are labeled J(SA), J(A), and J(B) and are from Roy and Robertson (1978). The pole labeled J2 is a secondary Jacobsville pole reported in the same study. The Chequamegon primary pole (C) and secondary pole (C2) are from this study.

samples is Keweenawan and not characteristic of the high-latitude Hadrynian track poles.

The secondary magnetization in Chequamegon samples has high blocking temperatures, high coercivities, and is removed during acid leaching at the same time as hematite pigment is removed. This magnetization thus resides in authigenic hematite. Since the Chequamegon beds are flat lying wherever they crop out, no fold test is possible to bracket the age of this secondary magnetization. This is unfortunate since the high-latitude poles that define the Hadrynian track are undated, and it is therefore of the utmost importance to determine whether these magnetizations are of recent origin or, as the supporters of the Hadrynian track maintain, are ancient magnetizations acquired during an episode of rapid Late Precambrian APW.

The timing of the acquisition of chemical remanence in red beds has been a subject of considerable debate. One particular controversy has centered around the nature of remanence in the Moenkopi Formation. One camp presents results of field tests suggesting that the Moenkopi carries a depositional remanence in detrital

specularite (Elston and Purucker 1979) and that a chemical remanence was acquired contemporaneously with deposition (Purucker *et al.* 1980). The opposing view, based on petrography and paleomagnetic analysis, maintains that remanence in the Moenkopi was acquired in a long and complicated process after deposition (Walker *et al.* 1981; Larson *et al.* 1982). Both sides in this debate give convincing arguments and the Moenkopi issue cannot be considered conclusively resolved at present. Extensive chemical demagnetization, without which the present study would not have been successful, may be helpful in resolving the Moenkopi debate.

Though questions about the value of red beds in reversal magnetostratigraphy remain, many red beds yield poles that, by comparison with studies of other rock types, appear to be suitable for APW studies. Roy and Park (1974) have used results from several studies to construct a three-phase model of the acquisition of magnetization in red beds. In this model, early, intermediate, and late stages occur in the initial remanence acquisition process. In some cases this process was shown to have occurred relatively early in the history of a rock. However, Turner (1979) listed a number of cases in which a stable magnetization was acquired long after deposition, and in a few cases in the present-day geomagnetic field. Given these considerations, secondary magnetizations in red beds must be used with caution. Secondary magnetizations are often acquired early, but they may be acquired at any time.

The high-latitude secondary poles from both the Jacobsville Sandstone (Roy and Robertson 1978) and the Chequamegon cluster so well around the present pole that it can be argued that they are of recent origin. Hamblin (1958), however, cited evidence based on an analysis of naturally leached areas that the red coloration in Jacobsville samples is an ancient feature. Models for the mechanism of a recent acquisition of the secondary remanence must take into account the fact that this hematite pigment and the secondary magnetization are in all probability related. Several possible models are: (1) that the secondary magnetization is a viscous magnetization (Biquand 1979); (2) that recent reorientation of the delicate hematite grains similar to the post-depositional reorientation of detrital grains in wet, fine-grained sediments (Kent 1973) may have occurred; and (3) that a recent chemical event involving local dissolution–reprecipitation occurred, resulting in a recent chemical magnetization. These models are speculative; however, whatever the mechanism, the permeable, poorly indurated Bayfield and Jacobsville Sandstones may be more susceptible to remagnetization than better lithified units (Turner 1979). Research on the complex problem of the origin of magnetization in red beds is still at an early stage, and further petrographic and experimental studies may eventually lead to satisfactory models.

Conclusions

This study underscores the importance of extensive chemical demagnetization in the study of red-bed paleomagnetism. Using this technique, two magnetizations have been isolated from Chequamegon Sandstone samples. One is a detrital magnetization residing, at least in part, in magnetite and yields a pole that falls among the late Keweenawan poles. The secondary remanence resides in hematite pigment and yields a pole close to the present north pole. Though the secondary remanence is not dated, we prefer the more conservative interpretation that this magnetization is of recent rather than of Hadrynian age. The Hadrynian track, which is based on secondary magnetizations, should not be considered a proven APW episode until well dated high-latitude poles from Upper Proterozoic rocks are found. Fold or conglomerate tests are crucial in this regard. Given these considerations, paleogeographic reconstructions based on the Hadrynian track must be regarded as tentative.

Acknowledgements

Daniel Frohling and David Gal helped with the paleomagnetic measurements. William C. Kelly helped with the reflected light microscopy. Dr. Robert Brander of the Apostle Islands National Lakeshore generously provided his time and knowledge during the fieldwork. The manuscript was reviewed by Jean Roy and Sherman Grommé. Field research was supported by grants from the Scott Turner Fund, Sigma Xi, and the Geological Society of America to Chad McCabe. Laboratory work was supported by the Division of Earth Sciences, the National Science Foundation (grants EAR-7614996 and EAR-8103031).

BIQUAND, D. 1979. On the stability of the viscous remanent magnetization in sedimentary rocks when submitted to AF demagnetization: new experimental results. Physics of the Earth and Planetary Interiors, **19**, pp. 209–227.

CHASE, C. G., and GILMER, T. H. 1973. Precambrian plate tectonics: the mid-continent gravity high. Earth and Planetary Science Letters, **21**, pp. 70–78.

DUBOIS, P. M. 1962. Paleomagnetism and correlation of Keweenawan rocks. Geological Society of Canada, Bulletin 71, 75 p.

ELMORE, R. D., and VAN DER VOO, R. 1982. Origin of hematite and its associated remanence in the Copper Harbor Conglomerate (Keweenawan), Upper Michigan. Journal of Geophysical Research. (In press.)

ELSTON, D. P., and GROMMÉ, C. S. 1974. Precambrian polar wandering from Unkar Group and Nankoweap Formation, eastern Grand Canyon, Arizona. *In* Geology of northern

Arizona, Part 1: Regional studies. Geological Society of America, Boulder, CO, pp. 96–117.

ELSTON, D. P., and PURUCKER, M. 1979. Detrital magnetization in red beds of the Moenkopi Formation. Journal of Geophysical Research, **84**, pp. 1653–1665.

FISHER, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**, pp. 295–305.

HALLS, H. C., and PALMER, H. C. 1981. Remagnetization in Keweenawan rocks. Part 2: Lava flows within the Copper Harbor Conglomerate, Michigan. Canadian Journal of Earth Sciences, **18**, pp. 1395–1408.

HALLS, H. C., and WEST, G. F. 1971. A seismic refraction survey in Lake Superior. Canadian Journal of Earth Sciences, **8**, pp. 610–630.

HAMBLIN, W. K. 1958. The Cambrian sandstones of northern Michigan. Michigan Geological Survey, Publication 51, 146 p.

HENRY, S. G. 1979. Chemical demagnetization: methods, procedures, and applications through vector analysis. Canadian Journal of Earth Sciences, **16**, pp. 1832–1841.

HENRY, S. G., MAUK, F. J., and VAN DER VOO, R. 1977. Paleomagnetism of the upper Keweenawan sediments: the Nonesuch Shale and Freda Sandstone. Canadian Journal of Earth Sciences, **14**, pp. 1128–1138.

IRVING, E., and MCGLYNN, J. C. 1976. Proterozoic magnetostratigraphy and the tectonic evolution of Laurentia. Philosophical Transactions of the Royal Society of London, Series A, **280**, pp. 433–469.

KENT, D. V. 1973. Post depositional remanent magnetization in a deep sea sediment. Nature, **246**, pp. 32–34.

LARSON, E. E., WALKER, T. R., PATTERSON, P. E., HOBLITT, R. P., and ROSENBAUM, J. G. 1982. Paleomagnetism of the Moenkopi Formation, Colorado Plateau: basis for long-term model of acquisition of chemical remanent magnetism in red beds. Journal of Geophysical Research, **87**, pp. 1081–1106.

MCWILLIAMS, M. O., and DUNLOP, D. J. 1975. Precambrian paleomagnetism: magnetizations reset by the Grenville Orogeny. Science, **190**, pp. 269–271.

MORRIS, W. A., and ROY, J. L. 1977. The discovery of the Hadrynian polar track and further study of the Grenville problem. Nature, **266**, pp. 689–692.

MYERS, W. D. 1971. The sedimentology and tectonic significance of the Bayfield Group (upper Keweenawan?), Wisconsin and Minnesota. Ph.D. thesis, the University of Wisconsin, Madison, WI, 269 p.

PALMER, H. C., HALLS, H. C., and PESONEN, L. J. 1981. Remagnetization in Keweenawan rocks. Part 1: Conglomerates. Canadian Journal of Earth Sciences, **18**, pp. 599–618.

PESONEN, L. J., and NEVANLINNA, H. 1981. Late Precambrian asymmetric reversals. Nature, **294**, pp. 436–439.

PIPER, J. D. A. 1981. Paleomagnetic study of the (Late Precambrian) West Greenland kimberlite–lamphrophyre suite: definition of the Hadrynian track. Physics of the Earth and Planetary Interiors, **27**, pp. 146–184.

PURUCKER, M. E., ELSTON, D. P., and SHOEMAKER, E. M. 1980. Early acquisition of characteristic magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona. Journal of Geophysical Research, **85**, pp. 997–1012.

ROBERTSON, W. A., and FAHRIG, W. F. 1971. The great Logan paleomagnetic loop—the polar wandering from Canadian Shield rocks during the Neohelikian era. Canadian Journal of Earth Sciences, **8**, pp. 1355–1372.

ROY, J. L., and PARK, J. K. 1974. The magnetization process of certain red beds: vector analysis of chemical and thermal results. Canadian Journal of Earth Sciences, **11**, pp. 437–476.

ROY, J. L., and Robertson, W. A. 1978. Paleomagnetism of the Jacobsville Formation and the apparent polar wander path for the interval −1100 to −670 m.y. for North America. Journal of Geophysical Research, **83**, pp. 1289–1304.

TURNER, P. 1979. The paleomagnetic evolution of continental red beds. Geological Magazine, **116**, pp. 289–301.

VAN HOUTEN, F. B. 1968. Iron oxides in red beds. Geological Society of America Bulletin, **79**, pp. 399–416.

VEROSUB, K. L. 1977. Depositional and postdepositional processes in the magnetization of sediments. Reviews of Geophysics and Space Physics, **15**, pp. 129–142.

WALKER, T. R., LARSON, E. E., and HOBLITT, R. P. 1981. Nature and origin of hematite in the Moenkopi Formation (Triassic), Colorado Plateau: a contribution to the origin of magnetism in red beds. Journal of Geophysical Research, **86**, pp. 317–333.

WATTS, D. R. 1981. Paleomagnetism of the Fond du Lac and the Eileen and Middle River sections with implications for Keweenawan tectonics and the Grenville problem. Canadian Journal of Earth Sciences, **18**, pp. 829–841.

ZIJDERVELD, J. D. A. 1967. A.C. demagnetization of rocks: analysis of results. *In* Methods in paleomagnetism. *Edited by* D. W. Collinson, K. M. Creer, and S. K. Runcorn. Elsevier, Amsterdam, The Netherlands, pp. 254–286.